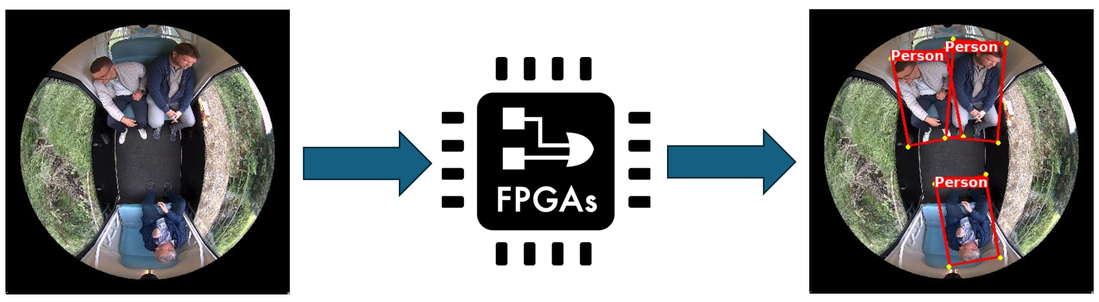
# FINN4YOLO: Unlocking Real-Time Fisheye Vision with FINN-Accelerated YOLO on FPGA

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## Motivation

- Fisheye cameras provide an ultra-wide field of view, making them attractive for autonomous robots, surveillance, and intelligent transportation systems.
- Fisheye image processing can benefit from edge deployment to reduce latency and bandwidth requirements while improving data privacy.
- Key challenges for real-time edge deployment include:
  - Fisheye cameras introduce geometric distortion in the images
  - Oriented object detection models have high computational cost
- **Goal: Enable efficient real-time oriented object detection on FPGAs for fisheye vision, particularly in moving platforms such as trains.**




## Contribution

- We develop FINN-compatible oriented bounding-box (OBB) detection architectures for FPGA deployment.
- We adapt YOLOv3n-tiny, YOLOv5n, and YOLOv8n to address FINN compilation and hardware constraints.
- We evaluate detection accuracy across four fisheye datasets and benchmark FPGA performance on ZCU104 and Alveo U250 platforms.

## Methodology and Key Ideas

- Development of customized quantized OBB detection heads for FINN-compatible FPGA deployment:
  - YOLOv3n-tiny-OBB: Extended output channels to predict an orientation angle for each anchor while retaining the original backbone and neck.
  - YOLOv5n-OBB: Integrated the OBB head and removed skip connections to simplify the computation graph for FINN compilation.
  - YOLOv8n-OBB: Replaced SiLU activations with ReLU to ensure FINN compatibility.
- Quantization-aware training (QAT) using standard YOLO augmentations at an input resolution of 416 × 416 × 3.
- The extended FINN-plus compiler is used, integrating custom operations into the streamlining and hardware-layer conversion steps to enable YOLO model compilation.
- Hardware-aware FPGA compilation with automatic FIFO sizing and a target clock period of 10 ns.
- On-board evaluation of the deployed models for accuracy, throughput, resource utilization, and power consumption.

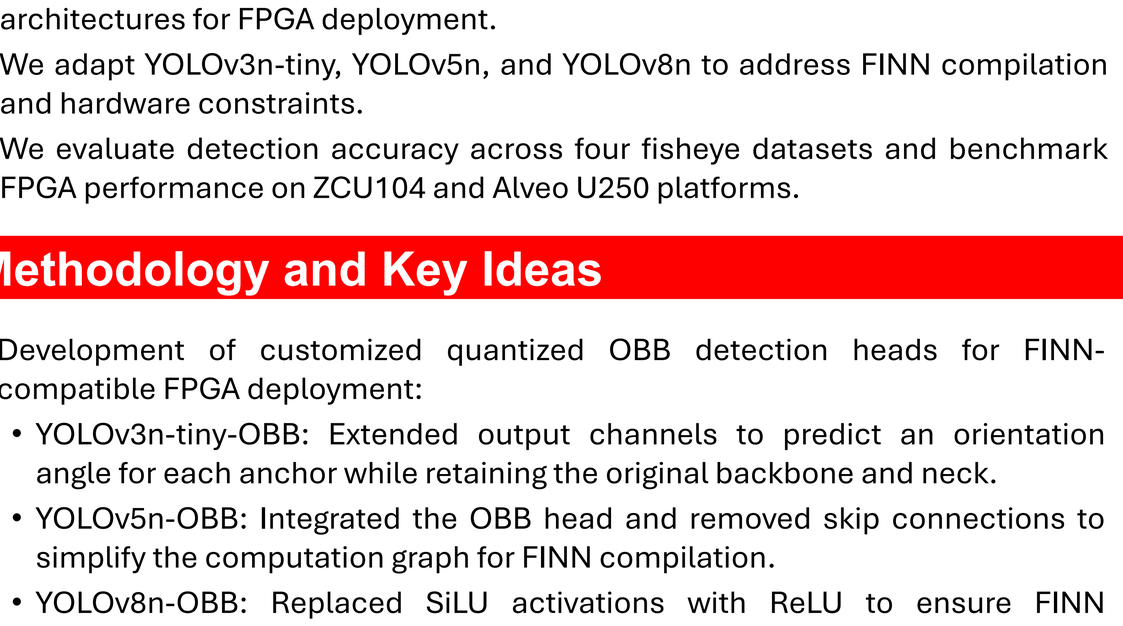



Figure 1: FINN-based workflow for training, compilation, FPGA deployment, and on-board evaluation of OBB YOLO models

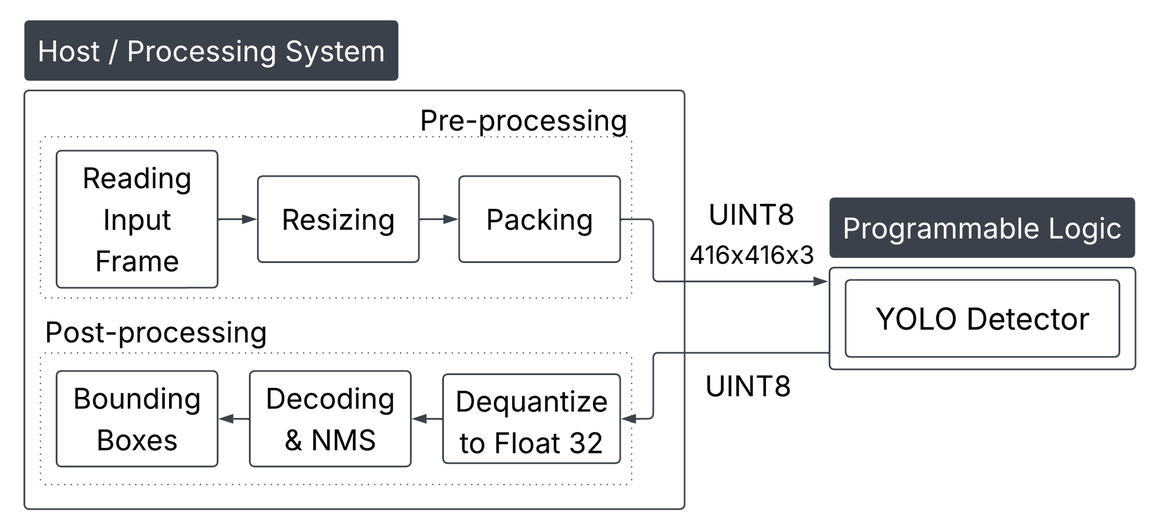



Figure 2: End-to-end FPGA inference pipeline comprising host-side preprocessing and post-processing and PL-based YOLO inference.

## Results

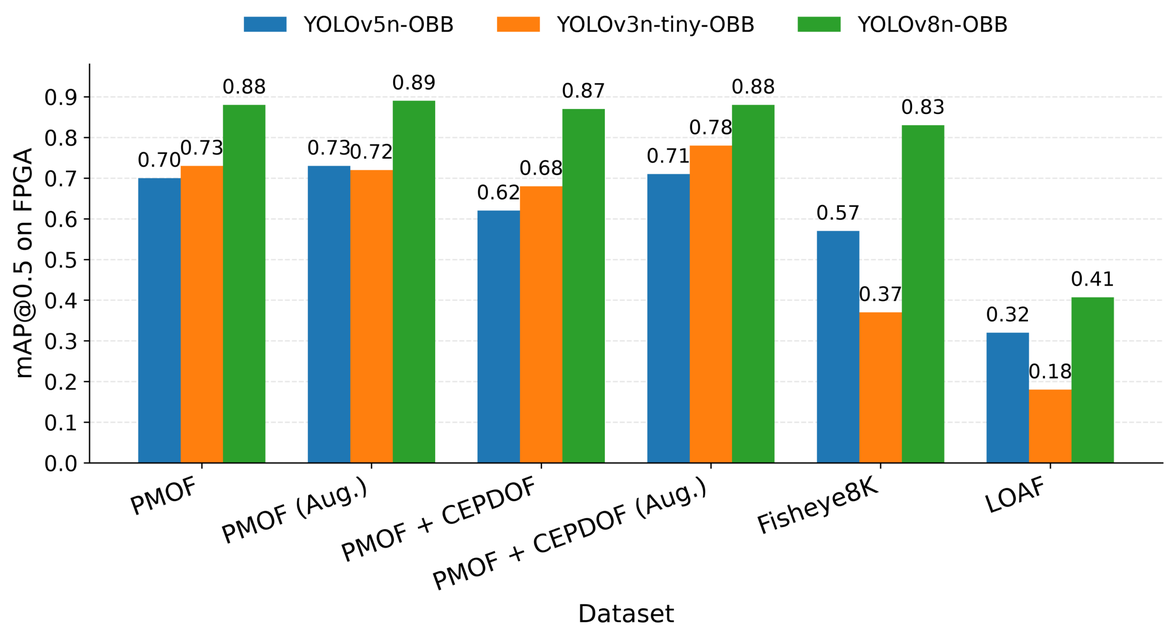



Figure 3: Detection accuracy of the models on FPGA across fisheye datasets.

| Model | Board | LUT | FF | BRAM | DSP | Power (W) |
|---|---|---|---|---|---|---|
| YOLOv3n-tiny-OBB | ZCU104 | 22.95 % | 10.12 % | 60.80 % | 1.79 % | 3.90 |
| YOLOv3n-tiny-OBB | U250 | 13.28 % | 7.57 % | 11.74 % | 0.18 % | 17.98 |
| YOLOv5n-OBB | U250 | 23.50 % | 14.15 % | 29.78 % | 20.05 % | 22.30 |
| YOLOv8n-OBB | U250 | 28.59 % | 16.75 % | 43.63 % | 17.94 % | 23.84 |

Table 1: Resource utilization and on-chip power consumption across models and FPGA platforms.

**The evaluated architectures demonstrate distinct accuracy–throughput–resource trade-offs, reaching up to 234 FPS on the Alveo U250.**

## Discussion

- YOLOv3n-tiny-OBB delivers the highest throughput: 234 FPS on U250 and 42 FPS on ZCU104, with 0.74 mAP@0.5 on PMOF.
- YOLOv5n-OBB provides a balanced design point (62 FPS, 0.71 mAP@0.5), while YOLOv8n-OBB favors accuracy at higher resource cost and lower throughput (30 FPS).
- Overall, the models expose clear accuracy–throughput–resource–power trade-offs, enabling application-specific FPGA deployment.

## Future Work

- Deployment of YOLOv11n-OBB and YOLO26n-OBB on FPGA using the FINN toolchain.
- Evaluate the performance of the deployed models across all six datasets in terms of latency, FPS, and on-chip power consumption.
- Explore resolution and quantization trade-offs for improved efficiency.

## References


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- Yang et al., 2023. Large-Scale Person Detection and Localization using Overhead Fisheye Cameras. Dataset
- Gochoo et al., 2023. FishEye8K: A Benchmark and Dataset for Fisheye Camera Object Detection. Dataset
- Duan et al., 2020. RAPiD: Rotation-Aware People Detection in Overhead Fisheye Images. Dataset


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